

Experiment 02 :

Output:

1. Attach complete program (with detailed comments explaining important steps) performing encryption and decryption of playfair cipher.
2. Display the key matrix created out of a given keyword.
3. Display the output of code for plaintext “the key is hidden under the door pad” with keyword “GUIDANCE” and its decryption.

Code:

```
# -----  
# Step 1 : Store all alphabets  
# -----  
alphabets = "ABCDEFGHIKLMNOPQRSTUVWXYZ" # J is omitted  
  
# -----  
# Step 2 : User Input for Keyword  
# -----  
keyword = input("Enter Keyword : ").upper()  
keyword = keyword.replace("J", "I")  
  
# -----  
# Step 3 : User Input for Plaintext  
# -----  
plaintext = input("Enter Plain Text : ").upper()  
  
# Remove spaces  
plaintext = plaintext.replace(" ", "")  
  
# Replace J with I  
plaintext = plaintext.replace("J", "I")  
  
# -----  
# Step 4 : Split Plaintext into Pairs  
# -----  
pairs = []  
  
index = 0  
  
while index < len(plaintext):  
    first = plaintext[index]  
  
    if index + 1 == len(plaintext):  
        second = "X"  
        index += 1  
  
    else:  
        second = plaintext[index + 1]  
  
        if first == second:  
            second = "X"  
            index += 1  
        else:  
            index += 2  
  
    pairs.append(first + second)  
  
print("\nPlaintext Pairs")  
print(pairs)
```

```
# -----  
# Step 5 : Generate 5x5 Matrix  
# -----  
matrix_string = ""  
  
# Add keyword letters  
for letter in keyword:  
    if letter not in matrix_string and letter in alphabets:  
        matrix_string += letter  
  
# Add remaining alphabets  
for letter in alphabets:  
    if letter not in matrix_string:  
        matrix_string += letter  
  
# Create Matrix  
matrix = []  
  
position = 0  
  
for row in range(5):  
    current_row = []  
  
    for column in range(5):  
        current_row.append(matrix_string[position])  
        position += 1  
  
    matrix.append(current_row)  
  
print("\nPlayfair Key Matrix\n")  
  
for row in matrix:  
    print(" ".join(row))  
  
# -----  
# Function : Find Position  
# -----  
def find_position(character):  
    for row in range(5):  
        for column in range(5):  
            if matrix[row][column] == character:  
                return row, column
```

```
# Step 6 : Encryption  
# -----  
encrypted_text = ""  
  
for pair in pairs:  
    first = pair[0]  
    second = pair[1]  
  
    row1, col1 = find_position(first)  
    row2, col2 = find_position(second)  
  
    # Same Row  
    if row1 == row2:  
        encrypted_text += matrix[row1][(col1 + 1) % 5]  
        encrypted_text += matrix[row2][(col2 + 1) % 5]  
  
    # Same Column  
    elif col1 == col2:  
        encrypted_text += matrix[(row1 + 1) % 5][col1]  
        encrypted_text += matrix[(row2 + 1) % 5][col2]
```

```
# Rectangle Rule
else:

    encrypted_text += matrix[row1][col2]
    encrypted_text += matrix[row2][col1]

print("\nEncrypted Text :")
print(encrypted_text)

# -----
# Prepare Ciphertext Pairs
# -----
cipher_pairs = []

for i in range(0, len(encrypted_text), 2):
    cipher_pairs.append(encrypted_text[i:i + 2])
```

```
# -----
# Step 7 : Decryption
# -----
decrypted_text = ""

for pair in cipher_pairs:

    first = pair[0]
    second = pair[1]

    row1, col1 = find_position(first)
    row2, col2 = find_position(second)

    # Same Row
    if row1 == row2:

        decrypted_text += matrix[row1][(col1 - 1) % 5]
        decrypted_text += matrix[row2][(col2 - 1) % 5]

    # Same Column
    elif col1 == col2:

        decrypted_text += matrix[(row1 - 1) % 5][col1]
        decrypted_text += matrix[(row2 - 1) % 5][col2]

    # Rectangle Rule
    else:

        decrypted_text += matrix[row1][col2]
        decrypted_text += matrix[row2][col1]

print("\nDecrypted Text :")
print(decrypted_text)
```

4. Display the output of code (encryption as well as decryption) for plaintext given in post experiment exercise.

Output:

```
Enter Keyword : GUIDANCE
Enter Plain Text : the key is hidden under the door pad

Plaintext Pairs
['TH', 'EK', 'EY', 'IS', 'HI', 'DX', 'DE', 'NU', 'ND', 'ER', 'TH', 'ED', 'OX', 'OR', 'PA', 'DX']

Playfair Key Matrix

G U I D A
N C E B F
H K L M O
P Q R S T
V W X Y Z

Encrypted Text :
POCLBXDRLGIYIBCGBGLXPOBILZLTTGIY

Decrypted Text :
THEKEYISHIDXDENUNDERTHEDOXORPADX
```

Post Experimental Exercise:

1. Encrypt message using playfair cipher : “The world of Cryptography”

Key: DECRYPTION [theoretical result and code output attached should match]

Output:

```
Enter Keyword : DECRYPTION
Enter Plain Text : The world of Cryptography

Plaintext Pairs
['TH', 'EW', 'OR', 'LD', 'OF', 'CR', 'YP', 'TO', 'GR', 'AP', 'HY']

Playfair Key Matrix

D E C R Y
P T I O N
A B F G H
J K L M Q
S U V W X

Encrypted Text :
NBRUGOJCIGRYDNINMOJAQN
```